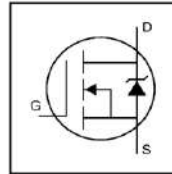


- Logic-Level Gate Drive
- Advanced Process Technology
- Dynamic dv/dt Rating
- 175°C Operating Temperature
- Fast Switching
- Fully Avalanche Rated

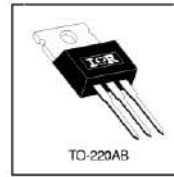


$$V_{DS} = 55V$$
$$R_{DS(on)} = 0.022\Omega$$
$$I_D = 47A$$

Description

Fifth Generation HEXFETs from International Rectifier utilize advanced processing techniques to achieve the lowest possible on-resistance per silicon area. This benefit, combined with the fast switching speed and ruggedized device design that HEXFET Power MOSFETs are well known for, provides the designer with an extremely efficient device for use in a wide variety of applications.

The TO-220 package is universally preferred for all commercial-industrial applications at power dissipation levels to approximately 50 watts. The low thermal resistance and low package cost of the TO-220 contribute to its wide acceptance throughout the industry.

**Absolute Maximum Ratings**

	Parameter	Max.	Units
I_D @ $T_C = 25^\circ\text{C}$	Continuous Drain Current, $V_{GS} = 10V$	47	A
I_D @ $T_C = 100^\circ\text{C}$	Continuous Drain Current, $V_{GS} = 10V$	33	A
I_{DM}	Pulsed Drain Current ^①	160	A
P_D @ $T_C = 25^\circ\text{C}$	Power Dissipation	110	W
	Linear Derating Factor	0.71	W/°C
V_{GS}	Gate-to-Source Voltage	± 16	V
E_{AS}	Single Pulse Avalanche Energy ^②	210	mJ
I_{AR}	Avalanche Current ^③	25	A
E_{ARR}	Repetitive Avalanche Energy ^④	11	mJ
dv/dt	Peak Diode Recovery dv/dt ^⑤	5.0	V/ns
T_J	Operating Junction and Storage Temperature Range	-55 to +175	°C
	Soldering Temperature, for 10 seconds	300 (1.6mm from case)	
	Mounting torque, 6-32 or M3 screw	10 lb•in (1.1N•m)	

Thermal Resistance

	Parameter	Min.	Typ.	Max.	Units
R_{JC}	Junction-to-Case	—	—	1.4	°C/W
R_{CS}	Case-to-Sink, Flat, Greased Surface	—	0.50	—	
R_{JA}	Junction-to-Ambient	—	—	62	

8/25/97

IRLZ44N

Electrical Characteristics @ $T_J = 25^\circ\text{C}$ (unless otherwise specified)

	Parameter	Min.	Typ.	Max.	Units	Conditions
$V_{(BR)DSS}$	Drain-to-Source Breakdown Voltage	55	—	—	V	$V_{GS} = 0V$, $I_D = 250\mu\text{A}$
$\Delta V_{(BR)DSS}/\Delta T_J$	Breakdown Voltage Temp. Coefficient	—	0.070	—	V/°C	Reference to 25°C , $I_D = 1\text{mA}$
$R_{DS(on)}$	Static Drain-to-Source On-Resistance	—	0.022	0.025	Ω	$V_{GS} = 10V$, $I_D = 25A$ ^①
		—	—	0.035		$V_{GS} = 5.0V$, $I_D = 25A$ ^②
		—	—	—		$V_{GS} = 4.0V$, $I_D = 21A$ ^③
$V_{GS(th)}$	Gate Threshold Voltage	1.0	—	2.0	V	$V_{DS} = V_{GS}$, $I_D = 250\mu\text{A}$
g_m	Forward Transconductance	21	—	—	S	$V_{DS} = 25V$, $I_D = 25A$
I_{OSS}	Drain-to-Source Leakage Current	—	25	—	μA	$V_{DS} = 55V$, $V_{GS} = 0V$
		—	250	—		$V_{DS} = 44V$, $V_{GS} = 0V$, $T_J = 150^\circ\text{C}$
I_{OSS}	Gate-to-Source Forward Leakage	—	100	—	nA	$V_{GS} = 16V$
		—	-100	—		$V_{GS} = -16V$
Q_g	Total Gate Charge	—	48	—	nC	$I_D = 25A$
Q_{gs}	Gate-to-Source Charge	—	8.6	—		$V_{DS} = 44V$
Q_{gd}	Gate-to-Drain ("Miller") Charge	—	25	—		$V_{GS} = 5.0V$, See Fig. 6 and 13 ^④
t_{don}	Turn-On Delay Time	—	11	—	ns	$V_{DD} = 28V$
t_r	Rise Time	—	84	—		$I_D = 25A$
$t_{d(off)}$	Turn-Off Delay Time	—	26	—		$R_D = 3.4\Omega$, $V_{GS} = 5.0V$
t_f	Fall Time	—	15	—		$R_D = 1.1\Omega$, See Fig. 10 ^⑤
L_D	Internal Drain Inductance	—	4.5	—	nH	Between lead, 6mm (0.25in.) from package and center of die contact
L_S	Internal Source Inductance	—	7.5	—		
C_{iss}	Input Capacitance	—	1700	—	pF	$V_{DS} = 0V$
C_{oss}	Output Capacitance	—	400	—		$V_{DS} = 25V$
C_{rss}	Reverse Transfer Capacitance	—	150	—		$f = 1\text{MHz}$, See Fig. 5

Source-Drain Ratings and Characteristics

	Parameter	Min.	Typ.	Max.	Units	Conditions
I_S	Continuous Source Current (Body Diode)	—	—	47	A	MOSFET symbol showing the integral reverse p-n junction diode.
I_{SM}	Pulsed Source Current (Body Diode) ^①	—	—	160	A	
V_{SD}	Diode Forward Voltage	—	—	1.3	V	$T_J = 25^\circ\text{C}$, $I_S = 25A$, $V_{GS} = 0V$ ^②
t_{rr}	Reverse Recovery Time	—	80	120	ns	$T_J = 25^\circ\text{C}$, $I_F = 25A$
Q_{rr}	Reverse Recovery Charge	—	210	320	nC	$dv/dt = 100A/\mu\text{s}$ ^③
t_{on}	Forward Turn-On Time	Intrinsic turn-on time is negligible (turn-on is dominated by $L_S + L_D$)				

Notes:

① Repetitive rating; pulse width limited by max. junction temperature. (See Fig. 11)
② $V_{DD} = 25V$, starting $T_J = 25^\circ\text{C}$, $L = 470\mu\text{H}$, $R_Q = 25\Omega$, $I_{AS} = 25A$. (See Figure 12)

③ $I_{SD} \leq 25A$, $di/dt \leq 270A/\mu\text{s}$, $V_{DD} \leq V_{(BR)DSS}$, $T_J \leq 175^\circ\text{C}$

④ Pulse width $\leq 300\mu\text{s}$; duty cycle $\leq 2\%$.